Plainly this is different than the Itō version of the integral, but in the cases we consider it is possible to reformulate Stratonovich integrals as Itō integrals. In the remainder of this book we will adopt the Itō form of stochastic integration.

Defined by Itō integration, $\mathrm{Y}(t) \stackrel{\text{def}}{=} \int_0^t g(s)\mathrm{dW}(s)$ can be thought of as a random process. In the case that g indicated here is a deterministic function, $\mathrm{Y}(t)$ is Gaussian in the sense that it is normally distributed for all time, as summarized in the following proposition.

Proposition 6.3 *Let g be a smooth deterministic function and $\mathrm{W}(t)$ a Wiener process. The stochastic integral $\mathrm{Y}(t) = \int_0^t g(s)\mathrm{dW}(s)$ is, for all times $t \geq 0$, a normally distributed random variable such that*

$$\mathbb{E}\left(\mathrm{Y}(t)\right) = 0,$$

$$\mathbb{E}\left(\mathrm{Y}^2(t)\right) = \sigma^2(t) = \int_0^t g(s)^2 \mathrm{d}s.$$

Proof See Exercise 2.

Finally, we observe that by calculating $\sigma^2(t_n)$ and generating random increments $R_n \sim \mathcal{N}(0, \sigma(t_n))$ we have a discrete representation of the continuous time stochastic process $\mathrm{Y}(t)$, effectively an exact stroboscopic representation. Also it is natural to view the stochastic integral as the solution of the stochastic differential equation (SDE) with zero initial condition:

$$\mathrm{dY}(t) = g(t)\mathrm{dW}(t), \qquad \mathrm{Y}(0) = 0. \tag{6.8}$$

This notion of stochastic integration carries over to stochastic integration with respect to more general categories of stochastic processes. Importantly, with appropriate regularity we have the familiar relation from calculus:

$$\int_0^t \mathrm{dY}(s) = \mathrm{Y}(t) - \mathrm{Y}(0).$$

6.3 Stochastic Differential Equations

In this section we briefly derive a more general stochastic differential equation as the limit of a biased random walk. The treatment here is intuitive, based on the concepts of limiting processes and stochastic integration introduced above. For a rigorous treatment of the material presented here, it is recommended to consult a reference such as the excellent book of Nelson [279] (which also contains a very lively historical discussion).